EXPOSYS DATA LABS | DATA SCIENCE INTERNSHIP PROGRAM

Diabetes Early-Risk Prediction Using Logistic Regression

Technical Report: Model Development, Validation and Selection

Prepared by	Souvik Chakraborty
Domain	Healthcare analytics: early detection of diabetes mellitus
Dataset	Diabetes prediction dataset, 100,000 patient records, 9 attributes
Model	Standardised numeric features + target-encoded categorical feature + Logistic Regression
Version	2.0 (supersedes the initial multi-algorithm report)

1. Executive Summary

Diabetes is a chronic disease that is far easier and cheaper to manage when it is identified early. This project builds a supervised machine-learning model that estimates whether a patient is diabetic from five routinely recorded clinical and lifestyle indicators: hypertension, heart disease, smoking history, body mass index (BMI) and HbA1c level.

The model is a single scikit-learn **Pipeline** that scales numeric features, target-encodes the categorical feature and fits a logistic regression classifier. It was trained on 80,000 records, assessed with stratified 5-fold cross-validation, and then confirmed once on a held-out test set of 20,000 records that was never used during training or tuning.

Key findings

- **Discrimination is strong and stable:** test ROC-AUC 0.9066 versus a cross-validated mean of 0.9043 +/- 0.0032.
- **Positive predictions are reliable:** precision of 0.8413 on the test set (about 84% of flagged patients are truly diabetic).
- **Sensitivity is the main weakness:** recall is 0.4118, so at the default 0.5 decision threshold the model detects roughly four in ten diabetic patients.
- **Selection outcome:** test and cross-validation results agree within the pre-defined stability rule, so the pipeline is selected as the baseline model, subject to threshold tuning and independent validation before any clinical use.

2. Problem Statement and Objectives

Predicting diabetes at an early stage allows patients to adjust their lifestyle and begin treatment before complications such as eye, kidney, nerve and cardiovascular damage appear. The objectives of this project are:

- Prepare a clean, reproducible modelling dataset from the raw patient records.
- Build a binary classifier that separates diabetic (1) from non-diabetic (0) patients.
- Evaluate the model with several clinically meaningful measures, not accuracy alone.
- Verify stability with stratified 5-fold cross-validation and an untouched test set, and document the model-selection decision.

3. Dataset

The raw data is supplied as a compressed CSV (*diabetes_prediction_dataset.csv*) and is read directly from the zip archive. It contains 100,000 records and 9 columns: gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level and the target *diabetes*. No record had a missing target value, and the target contains only the labels 0 and 1.

Sample of the modelling features (training split):

Hypertension	Heart disease	Smoking history	BMI	HbA1c
1	0	former	27.32	6.5
0	0	never	25.18	4.5
0	0	No Info	25.95	6.6
0	0	current	23.43	6.0
0	0	No Info	22.62	5.0

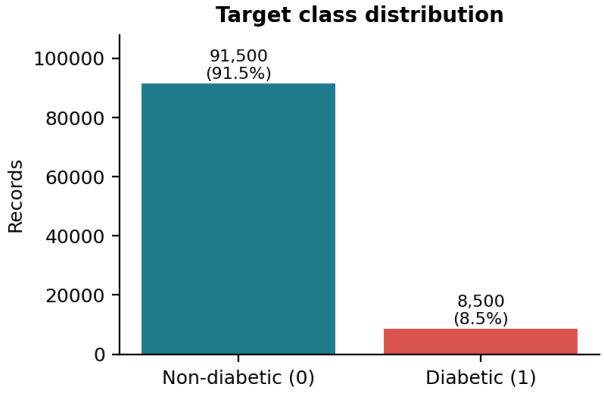


Figure 1. Diabetic patients form 8.5% of the data (class imbalance).

Class imbalance. Only 8,500 of 100,000 records (8.5%) are diabetic. A model that predicted "non-diabetic" for everyone would already score 91.5% accuracy. For this reason recall, precision, F1-score and ROC-AUC are reported alongside accuracy, and all data splits are stratified.

4. Feature Selection

Five predictors were retained for the model. Each has an established clinical association with diabetes risk:

Feature	Type	Clinical rationale
HbA1c level	Numeric	Reflects average blood sugar over about three months; the primary diagnostic marker of diabetes.
BMI	Numeric	Excess body weight is a major risk factor; risk rises sharply above a BMI of 30.
Hypertension	Binary (0/1)	Shares risk factors with diabetes; each condition tends to worsen the other.
Heart disease	Binary (0/1)	Diabetes and cardiovascular disease frequently occur together.
Smoking history	Categorical	Smoking is associated with insulin resistance and higher type 2 diabetes risk.

Gender, age and blood glucose level are present in the dataset but are not used in this version of the model. Age and blood glucose are known to be strongly related to diabetes, so evaluating their effect is listed under future work (Section 10).

5. Methodology

5.1 Workflow

Step	Description
1. Load and clean	Read CSV from the zip archive; drop rows with missing target; confirm binary target; remove duplicate columns.
2. Identify feature types	4 numeric features (hypertension, heart_disease, bmi, HbA1c_level) and 1 categorical feature (smoking_history).
3. Split	80% training (80,000) / 20% test (20,000), stratified on the target, random_state = 42.
4. Build pipeline	ColumnTransformer (preprocessing) followed by LogisticRegression, combined in one Pipeline object.
5. Baseline fit	Fit on the full training set; evaluate once on the held-out test set.
6. Cross-validation	Stratified 5-fold CV on the training set only; metrics reported for every fold.
7. Model selection	Compare test and CV results using a stability rule and print the decision with its reasoning.

5.2 Preprocessing and model

- **StandardScaler** on numeric features: centres each feature at mean 0 with unit variance, which helps the logistic-regression solver converge.
- **TargetEncoder** on smoking_history (smooth = auto, cv = 5): replaces each category by its smoothed mean diabetes rate. Internal cross-fitting prevents the encoder from seeing a row's own label, and unseen categories fall back to the global mean.
- **LogisticRegression** (max_iter = 2000, random_state = 42): an interpretable linear classifier that outputs a probability of diabetes, which suits clinical risk scoring.
- **Pipeline:** all preprocessing is learned from training folds only, so no information leaks from validation or test data.

5.3 Evaluation measures

Measure	Meaning in this context
Accuracy	Share of all patients classified correctly (inflated by class imbalance).
Precision	Of patients predicted diabetic, the share who truly are (controls false alarms).
Recall (sensitivity)	Of truly diabetic patients, the share the model detects (controls missed cases).
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Probability that a random diabetic patient is ranked above a random non-diabetic patient; independent of the threshold.

6. Tools and Environment

Category	Details
Language	Python (VS Code, virtual environment)
Libraries	pandas, NumPy, scikit-learn (TargetEncoder requires version 1.3 or later), zipfile; Matplotlib for visualisation
Data preparation	MS Excel for initial inspection of the dataset

7. Results

7.1 Baseline model: held-out test set (20,000 records)

Accuracy	Precision	Recall	F1-score	ROC-AUC
0.9434	0.8413	0.4118	0.5529	0.9066

7.2 Stratified 5-fold cross-validation (80,000 training records)

Fold	Accuracy	Precision	Recall	F1-score	ROC-AUC
Fold 1	0.9434	0.8503	0.4051	0.5488	0.9008
Fold 2	0.9434	0.8560	0.4022	0.5473	0.9040
Fold 3	0.9466	0.8719	0.4353	0.5807	0.9087
Fold 4	0.9421	0.8308	0.4007	0.5407	0.9019
Fold 5	0.9445	0.8471	0.4235	0.5647	0.9062
Mean +/- Std	**0.9440 +/- 0.0017**	**0.8512 +/- 0.0149**	**0.4134 +/- 0.0153**	**0.5564 +/- 0.0162**	**0.9043 +/- 0.0032**

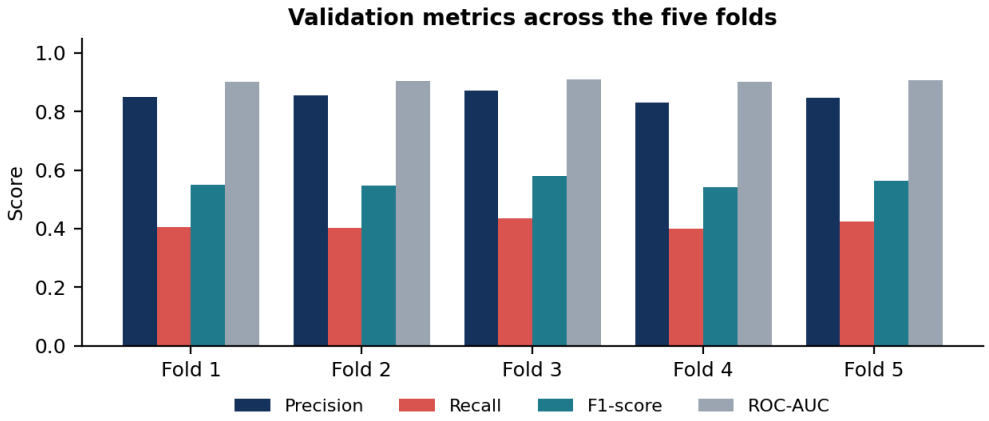


Figure 2. Precision, recall, F1-score and ROC-AUC for each cross-validation fold.

7.3 Interpretation

- **Consistency:** fold-to-fold variation is small (ROC-AUC standard deviation 0.0032), and test results sit within the cross-validation range for every metric. The model is not over-fitted and its performance is reproducible.
- **Ranking ability:** a ROC-AUC above 0.90 shows the model separates diabetic from non-diabetic patients well when the full probability range is considered.
- **Accuracy is misleading on its own:** 94.3% accuracy sounds high, but the 8.5% diabetic share means the baseline accuracy of a trivial model is 91.5%.
- **Recall gap:** recall of about 0.41 means many diabetic patients are classified as healthy at the default 0.5 threshold. Precision is high because the model only flags patients when it is very confident.

7.4 Approximate test-set confusion matrix

	Predicted non-diabetic	Predicted diabetic
Actual non-diabetic (about 18,300)	about 18,170 (TN)	about 130 (FP)
Actual diabetic (about 1,700)	about 1,000 (FN)	about 700 (TP)

Counts are estimated from the reported test metrics and the 8.5% prevalence; they are provided to illustrate the clinical meaning of the results.

8. Model Selection

The selection rule compares test performance with cross-validation performance. The model is approved when the test ROC-AUC and the test recall each lie within two cross-validation standard deviations of their cross-validated means.

Metric	Test value	CV mean	Gap	2 x CV std	Stable?
ROC-AUC	0.9066	0.9043	0.0023	0.0064	Yes
Recall	0.4118	0.4134	0.0016	0.0306	Yes

Decision: the Logistic Regression pipeline (StandardScaler + TargetEncoder + LogisticRegression), trained on the full training set, is selected as the baseline model. Both stability checks pass, fold-to-fold variation is low, and the model is simple, fast and explainable.

Note: the model is selected for *stability and generalisation*. It is not yet approved for clinical use because of the recall limitation discussed below.

9. Improvements over the Initial Version

Aspect	Initial report	This version
Categorical encoding	LabelEncoder on smoking_history (imposes a false numeric order)	TargetEncoder with smoothing and internal cross-fitting
Scaling	Only BMI standardised; fitted before the split	All numeric features scaled inside the pipeline, fitted on training data only
Data split	Random split, no seed, no stratification	Stratified 80/20 split with fixed seed (reproducible)
Evaluation	Accuracy only (single split)	Accuracy, precision, recall, F1, ROC-AUC; 5-fold CV plus held-out test
Model comparison	Logistic regression, decision tree and linear regression compared on accuracy and RMSE	Single validated pipeline with a documented selection rule
Leakage control	Scaler fitted on the full dataset	Pipeline refits all preprocessing on each training fold

The initial report concluded that linear regression was the best algorithm. That comparison was not valid: linear regression is a regression method evaluated by RMSE, which cannot be compared with the accuracy of a classifier, and accuracy alone hides the effect of class imbalance. This version replaces that conclusion with a like-for-like, multi-metric evaluation of a classification model.

10. Limitations, Clinical Considerations and Recommendations

10.1 Limitations

- **Low sensitivity:** about 59% of diabetic patients are missed at the 0.5 threshold. In a screening setting, missed cases are usually costlier than false alarms.
- **Single dataset:** results come from one dataset with one train/test split; no external or prospective validation has been performed.
- **Excluded predictors:** age, gender and blood glucose level were not used and may add predictive value.
- **Data quality:** smoking_history contains a large "No Info" group, and BMI shows many repeated values (for example 27.32) that suggest imputation in the source data.

- **Decision support only:** the model estimates risk and does not replace laboratory diagnosis or clinical judgement.

10.2 Recommendations

- **Tune the decision threshold** (for example from the precision-recall curve) or use *class_weight='balanced'* to raise recall to a clinically agreed target, accepting more false positives.
- **Add age and blood glucose level** and compare against the current five-feature model using the same CV protocol.
- **Benchmark non-linear models** (for example gradient boosting) against this baseline, keeping the same pipeline and selection rule.
- **Add calibration checks** (calibration curve, Brier score) so predicted probabilities can be trusted as risk scores.
- **Validate externally** on independent patient data and review with clinicians before deployment; monitor performance after release.
- **Maintain the code:** the installed scikit-learn version warns that TargetEncoder's *random_state* argument is deprecated (to be removed in version 1.11). Pass a shuffled StratifiedKFold object as *cv* instead.

11. Conclusion

A logistic-regression pipeline built from five clinical indicators discriminates well between diabetic and non-diabetic patients (ROC-AUC about 0.90) and performs consistently across cross-validation folds and an independent test set. Its precision is high, but its recall of about 0.41 is too low for a stand-alone screening tool. The model is therefore selected as a validated, interpretable baseline, and the next development cycle should focus on raising sensitivity through threshold tuning, additional predictors and independent validation before any use in a clinical workflow.

Appendix: Reproducibility Parameters

Item	Setting
Test size / split	0.20, stratified, random_state = 42
Training / test records	80,000 / 20,000
Cross-validation	StratifiedKFold, 5 splits, shuffle = True, random_state = 42
Numeric preprocessing	StandardScaler
Categorical preprocessing	TargetEncoder(smooth='auto', cv=5, random_state=42)
Classifier	LogisticRegression(max_iter=2000, random_state=42)
Scoring	accuracy, precision, recall, f1, roc_auc
Stability rule	|test - CV mean| <= 2 x CV std for ROC-AUC and recall